MEEG332 - Project 2: The Pinewood Drag

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Project Description: We are going to be using our new knowledge of lift and drag over objects and lifting surfaces, as well as computational fluid dynamics (CFD), to design vehicles for the Pinewood Derby. We are going to use an online CFD software to design derby car bodies that minimize drag and maximize lift in both 2D and 3D.

TASK 1: THEORETICAL MODEL

Diagram

A 2 dimensional diagram of the Pinewood derby track can be seen in Figure 1 below. It is dimensioned and positions 0, 1, and 2 are labeled, as well as the sections of the track named the "Hill" and the "Flat."

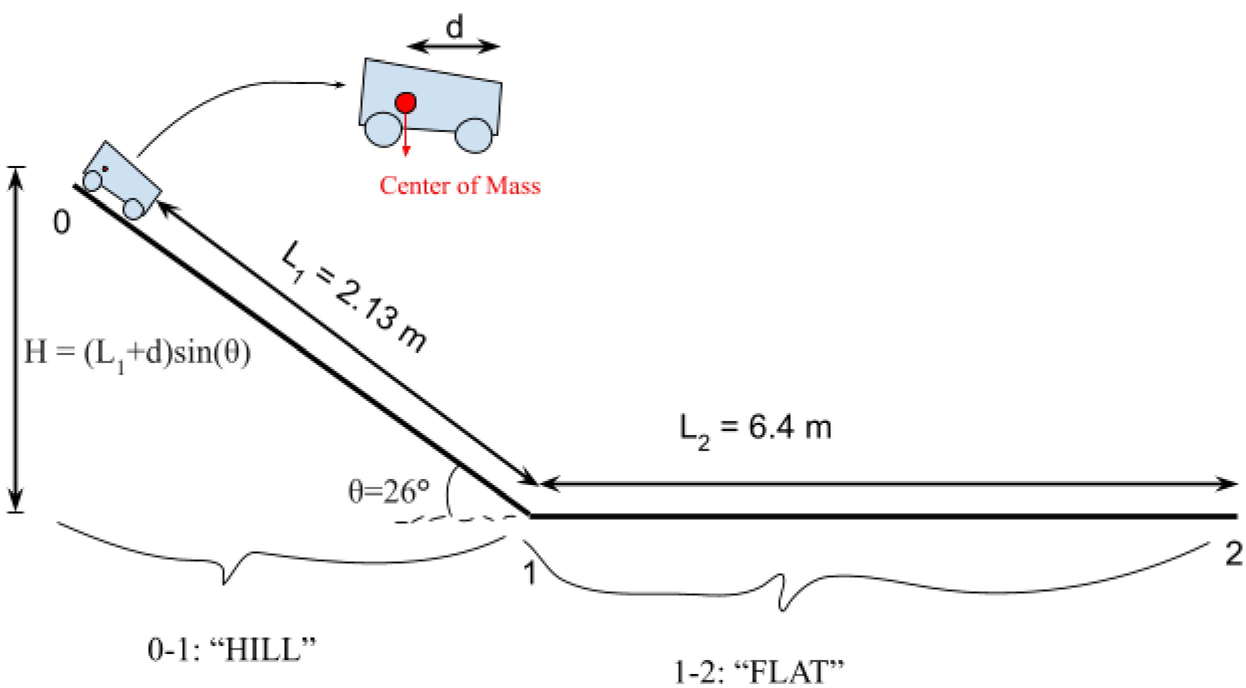

Figure 1: 2D diagram of the Pinewood Derby track\

The overall goal of the Pinewood Derby is to win the race. This is done by minimizing the time it takes to complete the race. Strategies can be employed for each section to reduce time.

Hill Section

The strategy for the "Hill" section is to maximize the amount of potential energy the car has at the beginning. Due to Conservation of Energy, the potential energy at the start of the race is directly correlated with the kinetic energy at the end of the race, shown by the relationship in Equation 1 below.

$$mgH = \frac{1}{2}mu_{car}^2 \quad [1]$$

where m is the mass of the car, g is the acceleration due to gravity, H is the height of the car's center of mass at the starting point of the track before the car is released, and u_{car} is the car's velocity. Height, as seen on Figure 1 above and Equation 2 below, can be calculated using:

$$H = (L_1 + d)sin(\theta) \quad [2]$$

where L_1 is the length of the "Hill" section, d is the distance from the front of the car to the center of mass of the car, and θ is the incline of the "Hill" section.

Flat Section

The strategy for the "Flat" section is to minimize the amount of drag on the car while maximizing the amount of lift. Still using the Law of Conservation of Energy, all frictional losses will detract from the starting energy. The first source of friction is called drag, which is caused by the car moving through the air and can be calculated by Equation 3 below.

$$F_D = \frac{1}{2}\rho u_{car}^2 AC_D \quad [3]$$

where ρ is air density, A is the surface area of the car, and C_D is the drag coefficient.

Maximizing lift force will decrease the amount of frictional forces caused by the wheel's contact with the axles. Lift can be calculated with Equation 4 below.

$$F_L = \frac{1}{2}\rho u_{car}^2 AC_L \quad [4]$$

where C_L is the lift coefficient. The reason why frictional forces are decreased due to lift is because friction is dependent on the normal force. Because the lift directly opposes the normal force caused by gravity, frictional forces are decreased. This relationship is shown by Equation 5 below.

$$F_F = 4r_{axle}\mu_f(F_w - F_L) \quad [5]$$

where r_{axle} is the radius of the axle, μ_f is the friction coefficient, and F_w is the force due to the weight of the car. $r_{axle} = 1.1\ mm$ and is the radius of the axles, and $\mu_f = 0.1$ for contact between graphite and steel.

Time

Finally, total time can be calculated to complete the race. This can be done by calculating the time it takes to complete each section. Time can be calculated in Equation 6 using the derivation below.

$$H = \frac{1}{2} g t_{0-1}^{2}$$

$$t_{0-1} = \sqrt{\frac{2H}{g}} \qquad [7]$$

where t_{0-1} is the time it takes to travel between positions 0 and 1.

For the flat section, Newton's Second Law can be used to equate the forces calculated in Equation 3 and 5 to the time derivative of velocity times mass to create a differential equation, shown in Equation 8.

$$m\frac{du_{car}}{dt} = - F_D - F_F$$

$$m\frac{du_{car}}{dt} = - \frac{1}{2}\rho u_{car}^{2} A C_D - 4 r_{axle} \mu_f \left(F_w + \frac{1}{2}\rho u_{car}^{2} A C_L\right) \qquad [8]$$

Solving this differential equation results in the time it takes to complete the "Flat" section, shown in Equation 9.

$$f(u_{car}) du_{car} = dt$$

$$t_{1-2} = g(u_{car}) \qquad [9]$$

The total time can be found by combining the times found in Equation 7 and 9. This formula is shown in Equation 10 below.

$$t_{tot} = t_{0-1} + t_{1-2} \qquad [10]$$

TASK 2: VEHICLE DESIGN

An isometric view of the design can be seen in Figure 2 below. The model of the car was taken from Professor Van Buren, and Pinewood properties were applied onto it for mass analysis.

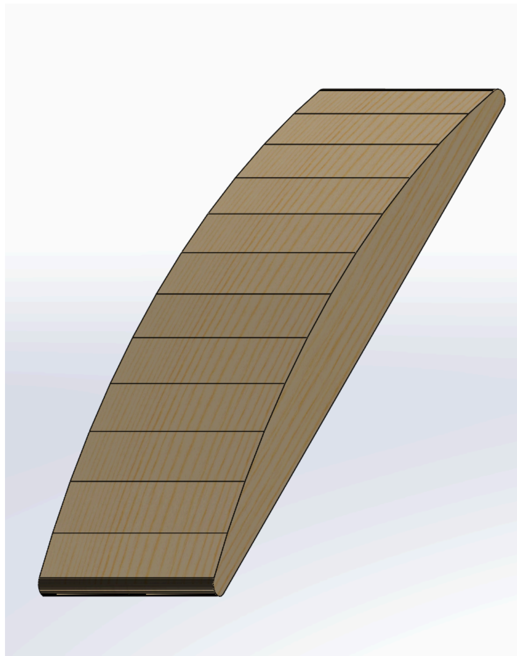

Figure 2: Isometric view of the car body in Solidworks

Figure 3 below shows a side profile of the car's body, showing where the wheel axles go, in red, as well as the center of mass.

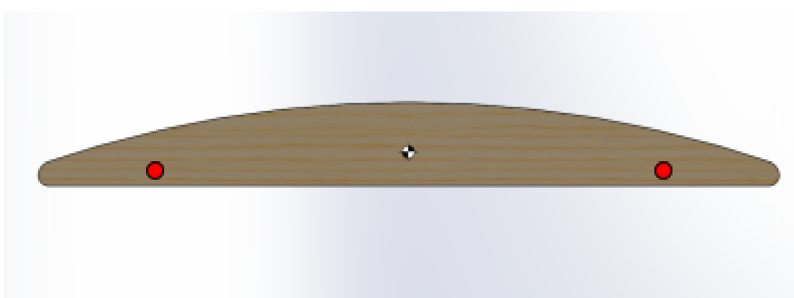

Figure 3: Side profile of the car's body, with the center of mass and the wheel axle locations

Although I did not design the vehicle, aerodynamic justification can still be made. Firstly, there are no sharp edges, with the ends being tapered to a tip. This will allow the air to flow around the surface and reduce drag. Also, the shape may also produce lift. The top surface is longer compared to the bottom surface which causes the airflow to travel faster over the top and slower underneath. As the air speeds up over the curved upper surface of the airfoil, it creates a region of lower pressure according to Bernoulli's principle. This lower pressure on the top surface compared to the bottom surface creates an upward force

TASK 3: VEHICLE PERFORMANCE

Figure 4 below shows the velocity profile of a 1.5 m/s airflow passing over the car simulated in Simscale. It includes a side and top view of the velocity profile. A constant velocity is observed before a decrease occurs as the car approaches the nose. This decrease is a result of the no-slip condition, which occurs when the stationary car interacts with the airflow. The air surrounding the vehicle exhibits a lower velocity compared to the airstreams farther away. Additionally, there are no eddies or reverse flows behind the vehicle, which is beneficial for minimizing drag.

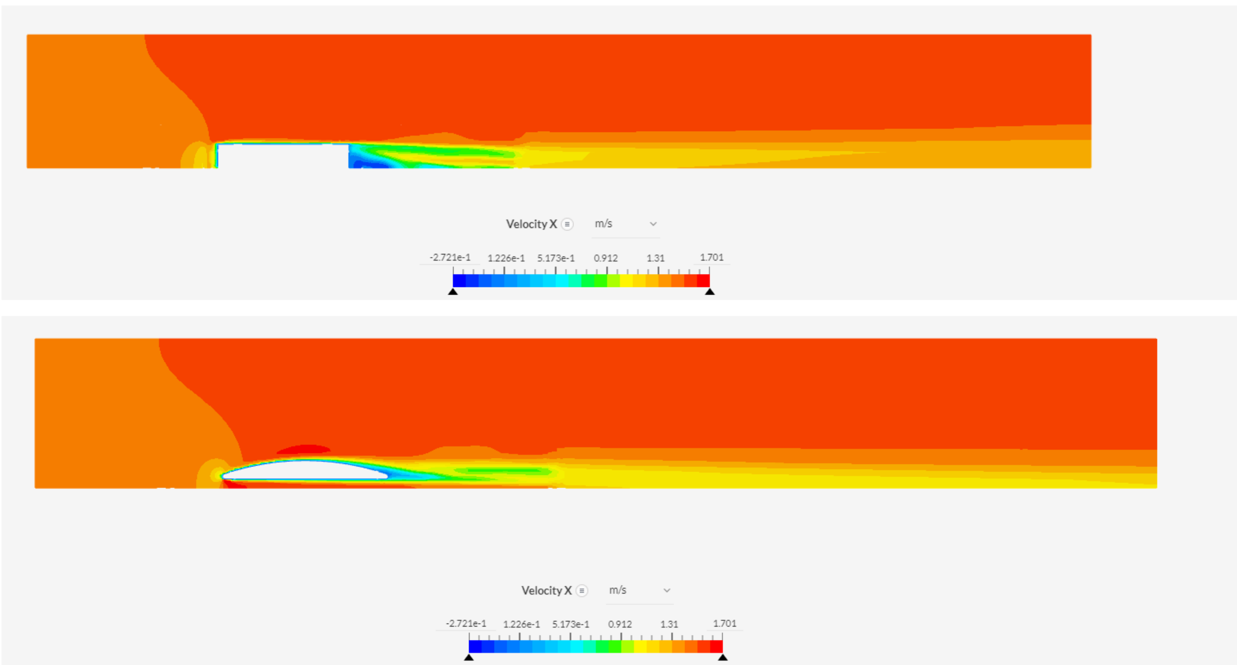

Figure 4: Side (top) and top (bottom) view of the velocity profile generated by Simscale

Figure 5 below shows the side and top view of the pressure flow field which displays a significant pressure concentration at the front of the car. This concentration arises from the abrupt velocity change as fast-moving air encounters the stationary vehicle. When the air comes into contact with the car's nose, it undergoes a sudden change in direction. Although the air surrounding the car experiences a slightly lower pressure compared to the rest of the air in the control volume, the pressure difference is negligible.

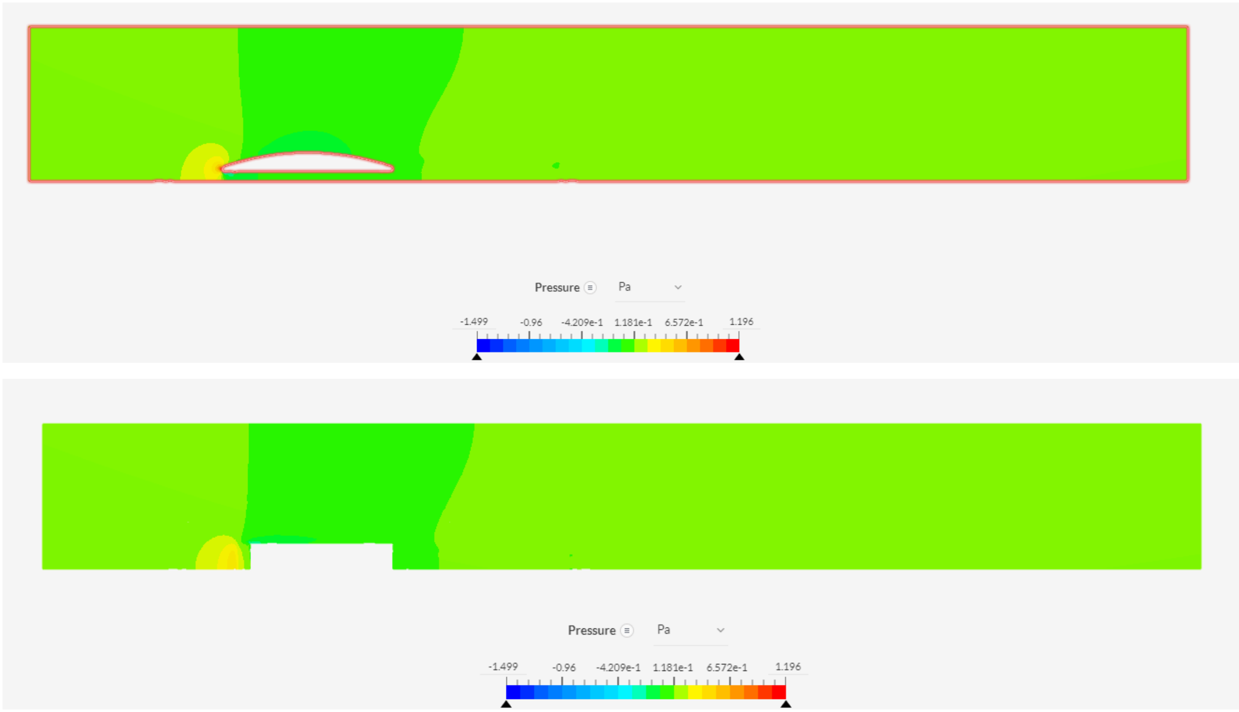

Figure 4: Side (top) and top (bottom) view of the pressure profile generated by Simscale

A plot of the forces was generated from Simscale. Average values were found for important values, which are tabulated below in Table 1, and are used to calculate the drag and lift forces.

Table 1: Average values for pressure and velocity, as well as the drag and lift forces

Property	**Value**
Average pressure force in the x-plane: F_{Px}	$1.13 * 10^{-4}\,N$
Average viscous force in the x-plane: F_{Vx}	$2.31 * 10^{-4}\,N$
Drag Force: $F_D = F_{Px} + F_{Vx}$	$3.44 * 10^{-4}\,N$
Average pressure force in the y-plane: F_{Py}	$4.31 * 10^{-4}\,N$
Average viscous force in the y-plane: F_{Vy}	$7.65 * 10^{-6}\,N$
Lift: $F_L = F_{Py} + F_{Vy}$	$4.38 * 10^{-4}\,N$

Rearranging Equations 3 and 4, we can solve for the lift and drag coefficient. We can use a value of 1.204 kg/m^3 for the value of ρ for air's density, and we can rearrange Equation 1 to find u_{car}. From Solidworks, the distance of the center of mass to the nose, d, is 88.9 mm, and the surface area of the car, A, is 25050 mm^2 or 0.02505 m^2. The calculations are shown below.

$$u_{car} = \sqrt{2gH} \quad [1']$$

$$u_{car} = \sqrt{2 * 9.8 * (2.13 + 0.00889)sin(26°)}$$

$$u_{car} = 4.29\ m/s$$

$$C_L = \frac{F_L}{\frac{1}{2}A\rho u_{car}^2} \quad [3']$$

$$C_L = 1.517 * 10^{-3}$$

$$C_D = \frac{F_D}{\frac{1}{2}A\rho u_{car}^2} \quad [4']$$

$$C_D = 1.931 * 10^{-3}$$

A summary of key aspects of the design can be seen in Table 2 below.

Table 2: Summary of car design

Parameter	**Value**
Center of Mass Distance from nose	$0.00889\ m$
Weight	$77.4\ g$
Volume	$1.436 * 10^{-4}\ m^3$
Surface Area	$0.02505\ m^2$
Lift Force	$4.38 * 10^{-4}\ N$
Drag Force	$3.44 * 10^{-4}\ N$
Lift Coefficient	$1.517 * 10^{-3}$
Drag Coefficient	$1.931 * 10^{-3}$